Generation of an Improved Precipitation Dataset from Multisource Information over the Tibetan Plateau

ZHONGKUN HONG,[a] ZHONGYING HAN,[a] XUEYING LI,[a] DI LONG,[a] GUOQIANG TANG,[a] AND JIANHUA WANG[b]

[a] *State Key Laboratory of Hydroscience and Engineering, Department of Hydraulic Engineering, Tsinghua University, Beijing, China*

[b] *State Key Laboratory of Simulation and Regulation of Water Cycles in River Basins, China Institute of Water Resources and Hydropower Research, Beijing, China*

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ABSTRACT: Precipitation over the Tibetan Plateau (TP), known as Asia's water tower, plays a critical role in regional water and energy cycles, largely affecting water availability for downstream countries. Rain gauges are indispensable in precipitation measurement, but are quite limited in the TP, which features complex terrain and a harsh environment. Satellite and reanalysis precipitation products can provide complementary information for ground-based measurements, particularly over large, poorly gauged areas. Here we optimally merged gauge, satellite, and reanalysis data by determining weights of various data sources using artificial neural networks (ANNs) and environmental variables including elevation, surface pressure, and wind speed. A Multi-Source Precipitation (MSP) dataset was generated at a daily time scale and a spatial resolution of 0.1° across the TP for the 1998–2017 period. The correlation coefficient (CC) of daily precipitation between the MSP and gauge observations was highest (0.74) and the root-mean-square error was the second lowest compared with four other satellite products, indicating the quality of the MSP and the effectiveness of the data merging approach. We further evaluated the hydrological utility of different precipitation products using a distributed hydrological model for the poorly gauged headwaters of the Yangtze and Yellow Rivers in the TP. The MSP achieved the best Nash–Sutcliffe efficiency coefficient (over 0.8) and CC (over 0.9) for daily streamflow simulations during 2004–14. In addition, the MSP performed best over the ungauged western TP based on multiple collocation evaluation. The merging method could be applicable to other data-scarce regions globally to provide high-quality precipitation data for hydrological research.

KEYWORDS: Precipitation; Hydrology; Remote sensing; Mountain meteorology; Neural networks

1. Introduction

The Tibetan Plateau (TP), known as the Earth's Third Pole, is the highest plateau in the world with a mean elevation of more than 4000 m. The TP and surrounding mountains are known as the water towers of Asia where headwaters of many major rivers in Asia reside, such as the Indus, Ganges, Brahmaputra, Yangtze, and Yellow Rivers (Huang et al. 2018a; Immerzeel et al. 2010). It has a large influence on global and regional climates and the water cycle (Han et al. 2021; Li and Long 2020; Li et al. 2019a,b; Nan et al. 2009; Wu et al. 2012). Research on the hydrological processes across the TP is of great importance for improving water resource management and mitigating natural hazards including floods and droughts in the TP and downstream countries (Huang et al. 2018b).

Precipitation plays a fundamental role in the water and energy cycles and shapes ecological landscapes and ecosystems, generating 55%–60% of annual runoff for all major rivers on the TP (Chen et al. 2013; Pang et al. 2017; Tang et al. 2018b; Tong et al. 2014b; Yang et al. 2004). There are currently four primary ways to estimate precipitation: 1) ground-based gauge observations, 2) ground-based radar remote sensing, 3) satellite remote sensing, and 4) atmospheric reanalysis models (Beck et al. 2017a). Gauge-based precipitation observations are generally considered most accurate and reliable. But there are almost no meteorological stations in the vast western and northern TP due to complex terrain and the harsh environment (Shen et al. 2014a; Y. Wang et al. 2019). The accuracy of the gridded precipitation products based on gauge data depends highly on the density of gauges and environmental conditions that are quite different in various regions (Chen et al. 2008; Hijmans et al. 2005). Therefore, it is necessary to combine satellite remote sensing and reanalysis data to further provide high-quality, high-resolution precipitation data.

Satellite remote sensing has been widely used in hydrometeorological applications, particularly for data-scarce areas (Han et al. 2020; Ma et al. 2020). Some popular global satellite remote sensing precipitation products were examined in this study, including TRMM Multisatellite Precipitation Analysis (TMPA) 3B42, Integrated Multisatellite Retrievals for GPM (IMERG) V06, Global Satellite Mapping of Precipitation (GSMaP), Climate Prediction Center morphing method (CMORPH), and Precipitation Estimation from Remotely Sensed Information Using Artificial Neural Networks–Climate Data Record (PERSIANN-CDR). Published studies show that TRMM-era data have a high correct detection rate of precipitation events in data-scarce areas such as high mountain

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Corresponding author: Di Long, dlong@tsinghua.edu.cn

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regions, and the near-real-time TRMM dataset has been successfully applied to flood monitoring in many studies (Hussain et al. 2018; Kim et al. 2017; Ning et al. 2017). GSMaP data, e.g., GSMaP_Gauge, that are adjusted by gauge data demonstrate comparable performance with gauge data, suggesting that GSMaP_Gauge could be used in hydrological research in the TP (Lu and Yong 2018). Some studies show that PERSIANN-CDR performs well in distinguishing rain from no-rain conditions and in estimating moderate rainfall (Alijanian et al. 2017; Katiraie-Boroujerdy et al. 2017). CMORPH shows reasonable spatiotemporal variability in precipitation in China, and has great potential in hydrological modeling and water resources management (Jiang et al. 2016; Shen et al. 2010). In addition, a global atmospheric reanalysis ERA5 that performs better than satellite-based data in winter and snow-covered areas, could further complement and improve the deficiencies of satellite data in these regions (Beck et al. 2019; Hersbach et al. 2020).

Satellite and reanalysis data have been used to investigate spatiotemporal changes in precipitation across the TP (Gao and Liu 2013; Ma et al. 2016; Maussion et al. 2014; Singh and Nakamura 2009; Wang and Zeng 2012; X. Wang et al. 2017; Yang et al. 2007; Yin et al. 2008; You et al. 2015), indicating the large potential of multisource data in hydrological research and applications over the vast data-scarce regions. However, satellite data generally overestimate light rainfall (e.g., 0–10 mm day^{-1}) and underestimate moderate and heavy rainfall (e.g., over 10 mm day^{-1}) in the TP (Gao and Liu 2013). Reanalysis data may not perform well in complex terrain, implying the necessity of considering environmental impacts (Liu et al. 2018). To provide high-quality and high-spatiotemporal-resolution precipitation data over the TP, satellite remote sensing and reanalysis data should be fully taken into account and multisource data need to be merged in a reasonable way.

As increasing precipitation datasets are produced, methods for data merging and correction are becoming critical to making use of various data sources. Multi-Source Weighted-Ensemble Precipitation (MSWEP) uses a weighted-ensemble method to merge high-quality precipitation data globally (Beck et al. 2017a, 2019). However, the effective spatial resolution of MSWEP is relatively coarse in the TP, since the reanalysis (~80–150 km) component is the dominating component (Beck et al. 2017a). The Global Precipitation Climatology Centre (GPCP) data combine high-accuracy microwave observations and more frequent geosynchronous infrared observations to provide high-spatiotemporal-resolution precipitation products (Adler et al. 2003, 2018). The NOAA CPC Unified gauge-based precipitation estimation optimally interpolates data from 30 000 stations, and improves the quality by considering topographic biases (Xie et al. 2007). There are also numerous studies that merge multisource precipitation data using various mathematical and/or statistical methods (Sapiano et al. 2008; Shen et al. 2014b; Y. Wang et al. 2019). These fusion methods mostly consider and correct for the impact of terrain on precipitation (e.g., the orographic effect), which should not be ignored for the TP. Furthermore, it is not enough to consider only the bias caused by the orographic effect over this region, because uncertainty in precipitation estimation in the TP is also affected by environmental factors (Shen et al. 2014a; Zhang et al. 2018).

Machine learning methods have reemerged in recent years, resulting in many hydrological applications including quantification of the impact of various factors on precipitation. Convolutional neural networks are used to improve the prediction of numerical models and provide more accurate precipitation estimates, which perform better than linear regression, nearest neighbor, random forest, or fully connected deep neural networks across the contiguous United States (Pan et al. 2019). Random forest, one of the machine learning approaches, outperforms the bilinear interpolation in downscaling precipitation data and can reproduce reasonable spatiotemporal patterns of precipitation across North America (He et al. 2016). In addition, the random forest–based merging procedure (RF-MEP) that combines gridded precipitation products performs well over Chile for 2000–16 (Baez-Villanueva et al. 2020). A recurrent neural network (RNN) model for simulating the hydrological response from various sources of rainfall was used to merge multiple precipitation sources for flash flood forecasting in Taiwan in China, indicating the potential of neural networks in merging multisource information (Chiang et al. 2007). There are comparisons between multivariate linear regression, artificial neural network, and spline interpolation applied in downscaling of satellite precipitation data and results show that neural networks have great potential (Sharifi et al. 2019).

The objective of this study is to develop a high-spatial-resolution and high-quality precipitation dataset using multisource information over the TP. The dataset spans from 1998 to 2017, with a spatial resolution of 0.1° and a temporal resolution at a daily time scale. First, we evaluated different precipitation products in the TP and selected appropriate precipitation products to be merged. Second, a precipitation dataset was generated by combining gauge, satellite remote sensing, and reanalysis data using artificial neural networks (ANNs) to determine weights of various data sources for the merging. Third, three approaches were used to fully evaluate and compare the merged dataset and different precipitation products: 1) independent gauge data for direct evaluation; 2) a distributed hydrological model [Coupled Routing and Excess Storage Model–Snow (CREST-Snow)] for indirect evaluation in headwaters of the Yangtze and Yellow Rivers; and 3) the multiple collocation (MC) method for evaluation in poorly gauged or ungauged areas (Pan et al. 2015). The merged dataset is termed as Multi-Source Precipitation (MSP) that entails high spatial resolution, high quality, and long time span, which is regarded among the best precipitation datasets for the entire TP, and could be valuable in climatic and hydrological research, water resource management, and natural hazards mitigation over the TP and its downstream areas.

2. Study area and data

a. Study area

The TP features the highest mean elevation and complex terrain in the world, with an elevation over 2500 m and a

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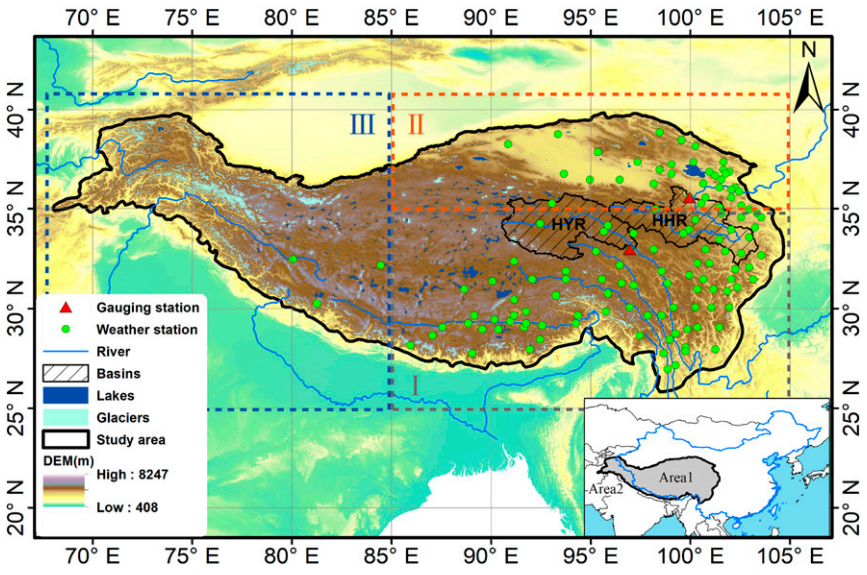


FIG. 1. Locations of the TP and surrounding regions, study basins (HYR and HHR are denoted by hatched areas), meteorological stations, and two gauging stations (i.e., Zhimenda and Tangnaihai denoted by red triangles), major rivers, lakes, glaciers, and elevations over the TP. The thick black solid line denotes the boundary of the TP with elevations higher than 2500 m. Area 1 in the inset denotes the part where China Gauge-based Daily Precipitation Analysis (CGDPA) dataset is available, and area 2 denotes the part beyond the domain of the CGDPA dataset but within the TP boundary.

sensitive region experiencing rapid climate change (Qin et al. 2009). It is located within the domain 68.25°–104.25°E and 26.25°–39.75°N in Central Asia with an area of ~2.5 million km^2 (X. Wang et al. 2018). Elevations of almost all meteorological stations over the TP are below 5000 m (the highest is 4910 m used in this study) and more than half are below 4000 m. As shown in Fig. 1, the number of meteorological stations is quite limited across the study region particularly in the western TP. Land surface temperatures of the TP vary greatly from region to region. The average temperature in the southeast can reach 20°C, while that in the northwest is mostly below 0°C. The TP was becoming warmer over the past decades at a warming rate of 0.3°C decade^{-1} (Wei and Fang 2013). The mean annual land surface temperature is about 3.1°C based on Moderate-Resolution Imaging Spectroradiometer (MODIS) products and likely to keep increasing in the twenty-first century (Chen et al. 2017; Qin et al. 2009). Annual precipitation decreases from the southeast (over 1000 mm) to the northwest (less than 100 mm) gradually and varies a lot on both sides of the Himalayas (X. Wang et al. 2018). Precipitation over the TP is mostly affected by the southwest monsoon, westerlies, and local convective activities (Romatschke and Houze 2011). The southeast region is much moister than the northwest region due to different water vapor contents (Li and Long 2020; X. Wang et al. 2018).

The TP was divided into three regions in this study with boundaries of 85°E and 35°N for further spatial analysis. Region I is the southeast region with relatively dense meteorological stations and heavy rainfall (X. Wang et al. 2018). Region II is the northeast region mainly including the Qaidam Basin, with less precipitation than Region I (Song et al. 2013). Region III is the western region with sparsely distributed gauges (X. Wang et al. 2018). Two headwater regions on the TP, i.e., the headwaters of the Yangtze (HYR) and Yellow (Huang) River (HHR) were chosen to evaluate precipitation products through hydrological modeling. The drainage area of the HYR basin is ~139 000 km^2, located within the domain 90.5°–97.3°E and 32.4°–35.8°N (P. Han et al. 2019). The outlet of the basin is the Zhimenda gauging station with in situ daily continuous discharges available for the 2004–14 period. The drainage area of the HHR basin is ~132 000 km^2, located within the domain 95.9°–103.4°E and 32.1°–36.0°N (Qin et al. 2017; T. Wang et al. 2018). The outlet of the basin is the Tangnaihai gauging station with in situ daily continuous discharges available during 2004–14.

b. *Precipitation data*

Precipitation data used in this study can be grouped into three categories: gauge, satellite remote sensing, and reanalysis data (Table 1). The gauge data comprise three gauge-based precipitation datasets, i.e., in situ point-based observations at meteorological stations from the China Meteorological Administration (CMA) and other two gauge-based gridded precipitation products. The CGDPA dataset is a gridded precipitation product generated by interpolating precipitation observations at rain gauges using the climatological optimal interpolation algorithm (Shen and Xiong 2016). WorldClim Version 2 (WorldClim2) is a dataset of spatially interpolated monthly climate data for global land areas based on weather station data from 1970 to 2000

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TABLE 1. Full names, spatiotemporal resolution, and sources of precipitation data used in MSP generated in this study. Temporal series of the data are not the entire temporal span of each product, but are the part used here.

	Short name	Full name and detail	Spatial resolution	Temporal resolution	Reference	Temporal coverage
Gauge	Gauge data	China Meteorological Station Observations	—	—	http://data.cma.cn/	1998–2017
	CGDPA	China gauge-based daily precipitation analysis	0.25°	Daily	Shen and Xiong (2016)	1998–2014
	WorldClim2	WorldClim version 2	0.083°	Monthly	Fick and Hijmans (2017)	—
Satellite	GSMaP	Global Satellite Mapping of Precipitation	0.1°	Hourly	Mega et al. (2019); Okamoto et al. (2005)	2001–17
	PERSIANN-CDR	Precipitation Estimation from Remotely Sensed Information Using Artificial Neural Networks (PERSIANN)–Climate Data Record (CDR)	0.25°	Daily	Ashouri et al. (2015)	1998–2017
	TMPA 3B42 V7	TRMM Multisatellite Precipitation Analysis (TMPA) 3B42	0.25°	3 hourly	Huffman et al. (2007)	1998–2000
	IMERG	Integrated Multisatellite Retrievals for GPM Final Run	0.1°	30 min	Huffman et al. (2015)	2001–17
	CMORPH	Climate Prediction Center morphing method	0.25°	Daily	Joyce et al. (2004)	1998–2017
Reanalysis	ERA5	Fifth generation of ECMWF atmospheric reanalyses of the global climate	0.25°	3 hourly	Hersbach et al. (2020)	1998–2017

(Fick and Hijmans 2017). Satellite remote sensing data include five products:

1) The gauge-adjusted Global Satellite Mapping of Precipitation (GSMaP_Gauge, termed as GSMaP hereafter) is a gridded precipitation dataset with resolution of 0.1° and 1 h, which is adjusted by CPC Unified (Mega et al. 2019; Okamoto et al. 2005).
2) Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks–Climate Data Record (PERSIANN-CDR) is a retrospective satellite-based precipitation dataset with resolution of 0.25° and 1 day (Ashouri et al. 2015).
3) Integrated Multisatellite Retrievals (IMERG) for GPM Final Run dataset was used from June 2000 to December 2017 in this study.
4) The TRMM Multisatellite Precipitation Analysis (TMPA) post-real-time research version 7 (3B42V7) with resolution of 0.25° and 3 h (Huffman et al. 2007) was included in MSP from January 1998 to May 2000 as a supplement to IMERG.
5) The Climate Prediction Center morphing method (CMORPH) is a satellite-based rainfall product with resolution of 0.25° and 1 day (Joyce et al. 2004).

The last one used in this study is ERA5. Compared to ERA-Interim that was superseded by the ERA5 reanalysis, ERA5 is improved greatly with resolution of 0.25° and 3-h (Hersbach et al. 2020). All gridded precipitation data were resampled to 0.1° using the nearest neighbor interpolation method.

To have a first-order understanding of the quality of precipitation data we collected, these products were evaluated with in situ precipitation measurements from 132 stations on the TP (Fig. 1) for the 2001–11 period. Evaluation results are shown using correlation coefficient (CC) between the daily precipitation estimation of these products and gauge observations in Fig. 2. CGDPA is highly consistent (highest CC values) with the gauge data at the corresponding grid cells, because it is generated by interpolating gauge data. GSMaP is corrected by NOAA CPC Unified Gauge-Based Analysis of Global Daily Precipitation dataset at the daily time scale. And IMERG is corrected by the Global Precipitation Climatology Centre (GPCC) product at the monthly time scale. The quality of them is relatively high in terms of CC values mostly greater than 0.6. PERSIANN-CDR is adjusted by the GPCP dataset, showing good performance in the southern TP in terms of CC values mostly close to 0.5. However, the quality of PERSIANN-CDR is relatively poor in terms of CC values lower than 0.3 over the eastern TP and even below 0.2 in the northwest region, which is attributed to the uncorrected nature of the product without gauge data.

In addition, the quality of another satellite precipitation product, CMORPH, is a little worse than IMERG but better than PERSIANN-CDR. It performs well in the eastern TP with CC values generally higher than 0.5, whereas it performs poorly in the central and western TP with CC values below 0.3 in the northwest TP. ERA5 is comparable to IMERG with CC values higher than 0.5 in almost all regions. Particularly, ERA5 performs relatively better than other satellite-based precipitation products in the western TP where meteorological stations are scarce (referring to section 2a and Fig. 2e). The

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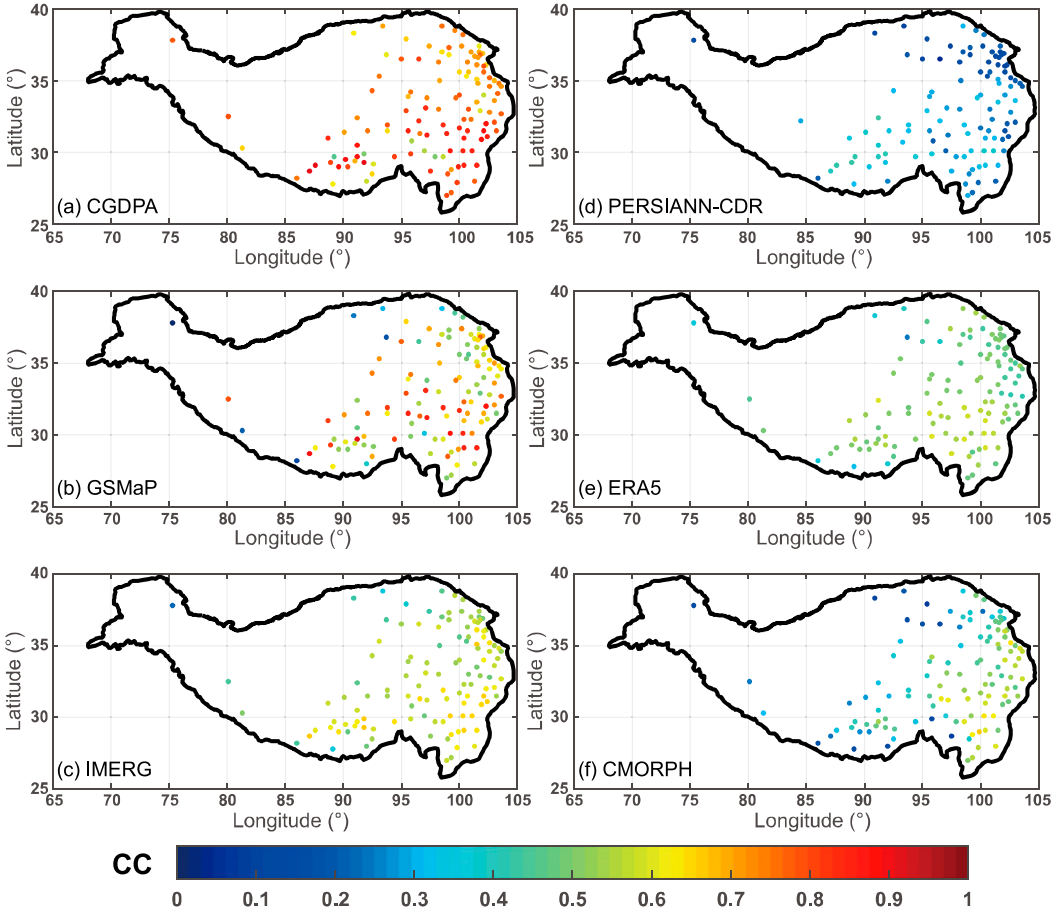


FIG. 2. Pearson CC distributions of daily precipitation data from various products during 2001–11. (a) CGDPA, (b) GSMaP, (c) IMERG, (d) PERSIANN-CDR, (e) ERA5, and (f) CMORPH. Dark blue indicates the lowest CC and dark red indicates the highest CC.

reanalysis data are expected to improve the estimation of precipitation in the western TP in this study.

To make a comparison among the MSP dataset and other satellite-based precipitation products using the MC method, we also used two other satellite precipitation products as independent references that were, however, not used in the merged dataset (MSP): 1) Climate Hazards Infrared Precipitation with Stations (CHIRPS) and 2) the Global Precipitation Climatology Product Version 3.1 (GPCP V3.1). CHIRPS is a satellite-based precipitation product with 0.05° spatial resolution which merges station data from five public data sources (i.e., GHCN monthly and daily, Global Surface Summary of the Day, GTS daily, Southern African Science Service Center for Climate Change and Adaptive Land Management) and other private datasets from various countries (Funk et al. 2015). GPCP V3.1 is a satellite-based precipitation product with a spatial resolution of 0.5°, fairly coarse compared with other satellite-based products. GPCP V3.1 is merged from satellite-only estimates and the GPCC gauge analyses product at the monthly time scale (Huffman et al. 2020, 2001). These two products have been widely used in data evaluation and analyses at various spatiotemporal scales (Bayissa et al. 2017; Beck et al. 2017b; Dinku et al. 2018; Gebremichael et al. 2005; Ma et al. 2009; Roca et al. 2019; Tan et al. 2015; G. Wang et al. 2017; Zambrano-Bigiarini et al. 2017).

c. *Auxiliary data*

Environmental variables considered in this study include elevation, land surface pressure, 2-m temperature, and land surface wind speed. The elevation data were derived from the digital elevation model (DEM) provided by NASA's Shuttle Radar Topography Mission (SRTM) with 90-m spatial resolution (Farr et al. 2007). Land surface pressure, 2-m temperature, and land surface wind speed were derived from ERA5 with 0.25° spatial resolution and daily temporal resolution (Hersbach et al. 2020). To match the 0.1° spatial resolution of the final precipitation product, the four related variables were all resampled to 0.1° spatial resolution by the liner interpolation method.

In addition to various precipitation products, forcing data of hydrological modeling performed in this study also include temperatures (land surface temperatures from MODIS and near surface air temperatures from the China Meteorological Administration) and potential evapotranspiration from the Famine Early Warning Systems Network (http://earlywarning.usgs.gov/fews). Snow water equivalent (SWE) estimated using a snow depth product

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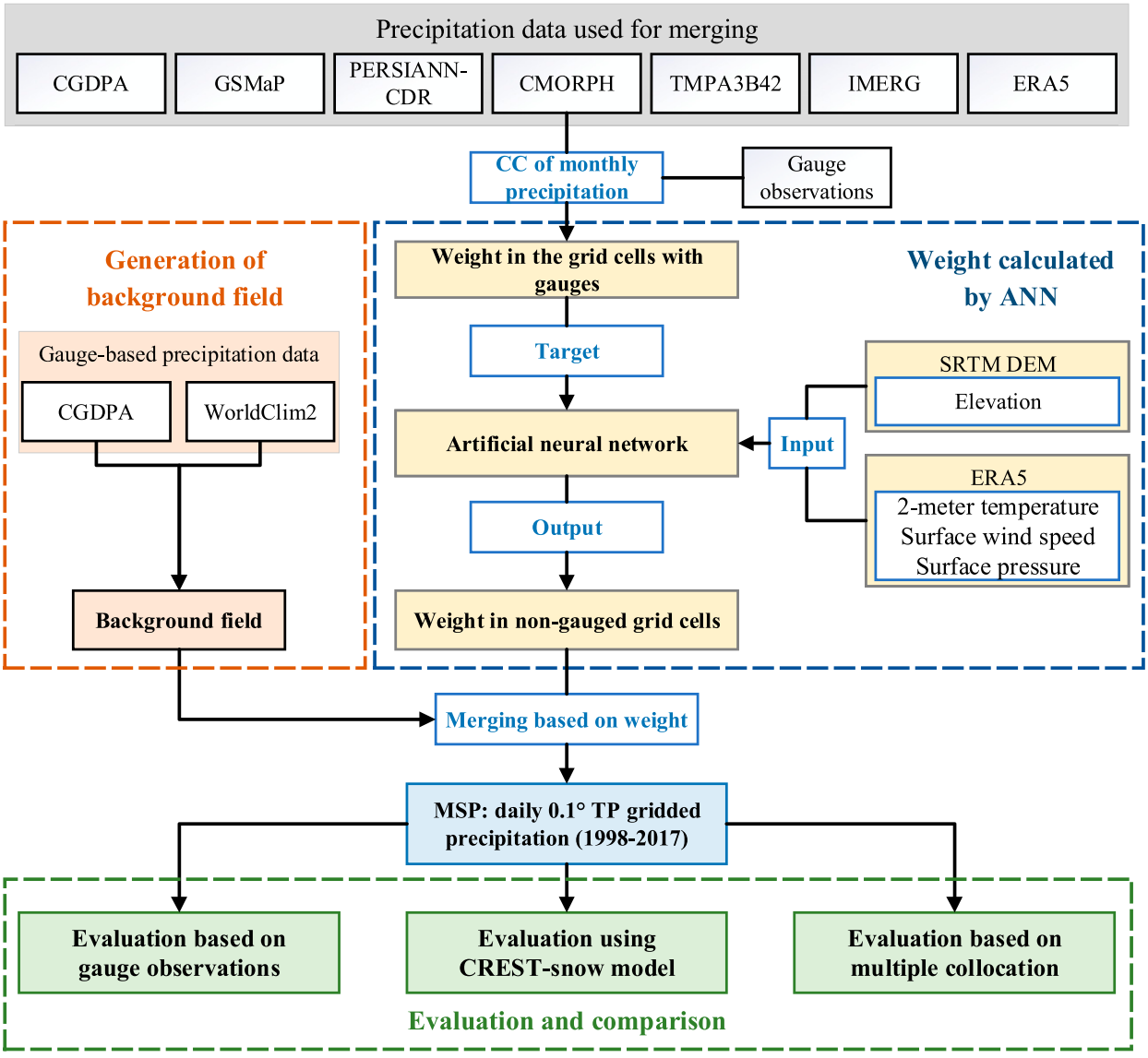


FIG. 3. Flowchart of generating MSP over the TP, including precipitation data preprocessing, construction of the precipitation background field, calculation of weights by ANN, and evaluation and comparison of all precipitation products.

obtained from the Environmental and Ecological Science Data Center for West China (http://westdc.westgis.ac.cn/) was used for calibrating and validating model parameters associated with snow accumulation and melting processes. In situ daily discharges at the Zhimenda and Tangnaihai gauging stations from January 2003 to December 2014 were obtained for warm up, calibration, and validation purposes for the hydrological model associated with runoff generation and routing (Chen et al. 2017).

3. Methodology

a. Framework

The framework of this study includes the following five major steps: 1) combining two gauge-based products (CGDPA and WorldClim2) as the background field for merging various precipitation products; 2) calculating CC values of various precipitation products with gauge data at the monthly scale to determine coefficients of determination in grid cells with stations; 3) determining weights in grid cells without stations by the ANN that incorporates environmental variables including elevation, surface pressure, wind speed, and 2-m temperature; 4) merging multisource precipitation products and generating the MSP dataset for the 1998–2017 period at a daily time scale and a spatial resolution of 0.1° across the TP; and 5) fully evaluating all precipitation products and MSP using independent gauge observations, hydrological modeling, and the MC method (Fig. 3).

b. Background field

A background field is the basis for precipitation data merging. Precipitation data from different sources are weighted into the

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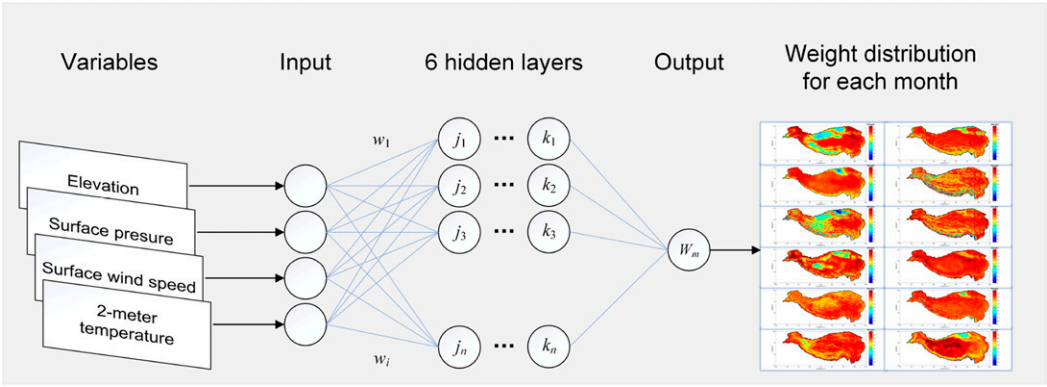


FIG. 4. ANN flowchart, including four environmental variables, input data, hidden layers, output data, and weight maps in 12 months.

background field that reflects climate characteristics across the TP. Therefore, the background field determines the overall quality of the final product. In this study, long-term average monthly precipitation of CGDPA and WorldClim2 was used to generate the background field. According to the first-order comparison (referring to section 2b), CGDPA in grid cells with weather stations is almost identical to the gauge data and is considered more accurate than WorldClim2. Given the different time spans (i.e., 1970–2000 for WorldClim2 and 1998–2014 for CGDPA) and data generation methods between these two products, further correction should be considered to combine them for determining the background field. For area 1 (Fig. 1), which belongs to the domain of the CGDPA dataset, long-term monthly precipitation of CGDPA was used as the background field. For area 2, which is the part beyond the domain of the CGDPA dataset, monthly precipitation of WorldClim2 was used as a substitute. Equation (1) was used to correct for the bias between the CGDPA and WorldClim2 precipitation estimates for area 2:

$$P_{b2} = \frac{\overline{P}_{C1}}{\overline{P}_{W1}} \times P_{W2}, \tag{1}$$

where P_{b2} is the monthly precipitation background field for area 2 (mm); P_{C1} is the monthly precipitation estimated from CGDPA for area 1 (mm); P_{W1} and P_{W2} are monthly precipitation estimated from WorldClim2 for area 1 and area 2, respectively (mm); and an overbar donates the mean precipitation during the time spans (1970–2000 for WorldClim2 and 1998–2014 for CGDPA). In this way, the monthly background field for the entire TP can be determined.

c. Basic weight calculation

Basic weights for different precipitation products should be first determined and then used for relative weight calculation (see section 3d). The basic weight is defined as the coefficient of determination between precipitation estimated from different products and rain gauge observations at the monthly time scale during 2001–14 (the common temporal coverage of all products in this study). For grid cells where gauges are located (120 grid cells on the TP), the basic weights can be directly estimated. Particularly, if higher weights result from the availability of gauges for a product, it does not mean that the product is better when there is no gauge to correct for the product. In fact, higher weights in grid cells with gauges do not guarantee that the product will do better where there is no gauge. Therefore, the ANN method and environmental variables are used to determine the basic weights for most grid cells without gauges.

This study built a back-propagation (BP) neural network, one of ANN methods to determine basic weights of various data sources (Fig. 4). The BP neural network has the nonlinear adaptive information processing ability to capture the complex relationship between input and output of training datasets (ASCE Task Committee on Application of Artificial Neural Networks in Hydrology 2000). There were 120 grid cells with gauge observations selected as the training datasets, and 10% of them were set as the validation data. We used TRAINLM as the training function, and set six hidden layers for trials. The input layers included coefficients of determination calculated from grid cells with gauge observations and four environmental variables (i.e., elevation, land surface wind speed, 2-m air temperature, and surface pressure), all of which were estimated at the monthly time scale. Finally, for each precipitation product, we can derive basic weight maps for each month.

The basic weight for each grid cell should be within the scale of 0–1. For a few grid cells having abnormal weights caused by the randomness of ANN, we used the median weight of 10 neighboring grid cells as a substitute, which is similar to the method used in MSWEP (Beck et al. 2017a). Although this may not be highly accurate, the uncertainty is negligible because the number of grid cells with abnormal weights is very small.

d. Merging data

According to the basic weight, precipitation anomalies estimated from different products were used to generate the MSP dataset based on the background field. Equation (2) was used to convert the basic weight into the relative weight for each product, and Eq. (3) was used to merge data based on relative weights and to generate the MSP dataset at the daily time scale:

$$rw_i = \frac{w_i}{\sum_{i=1}^{n} w_i}, \tag{2}$$

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TABLE 2. Statistical metrics used in evaluating precipitation products based on gauge-based observations. The term P_g is the daily precipitation observations at a meteorological station; P is the daily precipitation of various products estimated in the grid cell where the station is located; and an overbar donates the mean; cov is the covariance and σ is the standard deviation; N is the number of data pairs; F is the number of days that the precipitation product falsely detects precipitation events that do not occur in reality; H is the number of days that the precipitation product can correctly detect precipitation events; and M is the number of days that the precipitation product cannot detect precipitation events.

	Performance score	Score symbol	Equation
Continuous scores	Pearson's correlation coefficient	CC	$\mathrm{CC}=\frac{\mathrm{cov}(P,\,P_g)}{\sigma(P)\sigma(P_g)}$
	Root-mean-square error	RMSE	$\mathrm{RMSE}=\sqrt{\frac{\sum(P-P_g)^2}{N}}$
	Temporal variability ratio	STDRATIO	$\mathrm{STDRATIO}=\frac{\sigma(P)}{\sigma(P_g)}$
	BIAS	BIAS	$\mathrm{BIAS}=\frac{\overline{P}-\overline{P_g}}{\overline{P_g}}$
	Normalized root-mean-square error	NRMSE	$\mathrm{NRMSE}=\frac{\mathrm{RMSE}}{P_{g,\max}-P_{g,\min}}$
Categorical scores	False alarm ratio	FAR	$\mathrm{FAR}=\frac{F}{H+F}$
	Probability of detection	POD	$\mathrm{POD}=\frac{H}{H+M}$
	Critical success index	CSI	$\mathrm{CSI}=\frac{H}{H+F+M}$

where rw_i is the calculated relative weight for each product at the monthly time scale; w_i is the corresponding basic weight calculated from the ANN; and n is the number of precipitation products which are merged in a given month:

$$P_{\mathrm{MSP}}=\frac{P_b}{D_m}+\sum_{i=1}^{n} d_i\times rw_i, \tag{3}$$

where P_{MSP} is the generated precipitation estimates at the daily time scale; P_b is the precipitation background field, and D_m is the number of days in a given month; d_i is the daily precipitation anomaly in a month for each product, and n is the number of precipitation products which are merged; and rw_i is the relative weight for each product calculated by Eq. (2).

In most cases, precipitation values calculated by merging different precipitation products are larger than 0, which could result in the underestimation of the number of days without precipitation events. Therefore, the merged precipitation value is set to 0, if the sum of relative weights of precipitation products that detect no precipitation can reach 0.5.

TABLE 3. Statistical metrics used in evaluating the simulated discharge R against the observed discharge R_g during the time period T. An overbar donates the mean during evaluation periods; N is the number of data pairs; cov is the covariance, and σ is the standard deviation. For the OI calculation formula, E_i is one of the evaluation indices (i.e., NSE, LogNSE, CC, BIAS, and RMSE), and rank(E_i) indicates the score corresponding to the ranking of the indicator in each precipitation dataset, which equals 1, 0.8, 0.6, and 0.4 for the first, second, third, and fourth ranking. For example, if a precipitation product has CC and LogNSE ranking the first, NSE and RMSE ranking the second, and BIAS ranking the third among all products, the OI can be calculation by (1 + 1 + 0.8 + 0.8 +0.6)/5.

Performance score	Score symbol	Equation
Nash–Sutcliffe efficiency coefficient	NSE	$\mathrm{NSE}=1-\frac{\sum(R_g-R)^2}{\sum(R_g-\overline{R_g})^2}$
Logarithm Nash–Sutcliffe efficiency coefficient	LogNSE	$\mathrm{LogNSE}=1-\frac{\sum(\log R_g-\log R)^2}{\sum(\log R_g-\overline{\log R_g})^2}$
Pearson's correlation coefficient	CC	$\mathrm{CC}=\frac{\mathrm{cov}(R,\,R_g)}{\sigma(R)\sigma(R_g)}$
Bias	BIAS	$\mathrm{BIAS}=\frac{\sum(R-R_g)}{N}$
Root-mean-square error	RMSE	$\mathrm{RMSE}=\sqrt{\frac{\sum(R-R_g)^2}{N}}$
Overall index	OI	$\mathrm{OI}=\sum_{i=1}^{5}\mathrm{rank}(E_i)/5$

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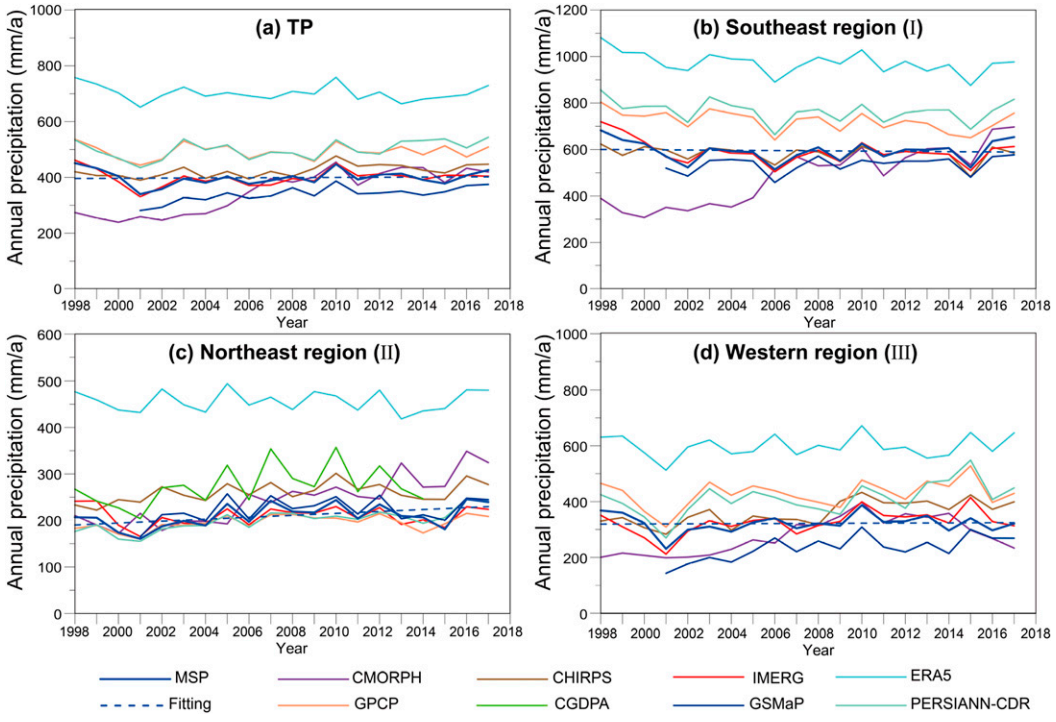


FIG. 5. Annual precipitation estimated from various products over the (a) TP, (b) southeast region (Region I), (c) northeast region (Region II), and (d) western region (Region III). Solid lines show annual precipitation estimated from different precipitation products, and the dash line in each plot shows the precipitation trend based on the MSP dataset.

e. Evaluation methods

1) EVALUATION USING GAUGE OBSERVATIONS

Precipitation observations from 114 meteorological stations across the TP for the 2015–17 period, independent of various precipitation products, were used to evaluate precipitation estimates at the daily time scale. Seven statistical metrics were used, including: 1) Pearson correlation coefficient (CC), 2) root-mean-square error (RMSE), 3) bias (BIAS), and 4) standard deviation ratio (STDRATIO), 5) false alarm rate (FAR), 6) probability of detection (POD), and 7) critical success index (CSI). Equations for these performance metrics are given in Table 2. To further analyze the MSP in different seasons, the normalized root-mean-square error (NRMSE) is also considered.

2) INDIRECT EVALUATION USING THE CREST-SNOW MODEL

Accuracy of precipitation estimates can be indirectly evaluated by comparing their model performance as the forcing data (Beck et al. 2017a; Jiang et al. 2016; Lu and Yong 2018). Using the same hydrological model with different precipitation estimates as the forcing data, performance of simulated discharge should be consistent with the quality of precipitation products, i.e., accurate precipitation estimates are likely to result in good performance of simulated discharge. The CREST-Snow model was used in this study to make indirect evaluation of different precipitation products. The original version of CREST-Snow, CREST V2.1, was a distributed hydrological model developed by the University of Oklahoma and NASA SERVIR (Wang et al. 2011). CREST-Snow was developed by combining a groundwater module and a snowmelt module to CREST V2.1 to improve its applicability in cryospheric regions such as the TP (Chen et al. 2017; Du and Long 2018). The CREST-Snow model has been used in discharge estimation and related hydrological applications over the TP, achieving promising results (P. Han et al. 2019; Z. Han et al. 2019; Han et al. 2020; Huang et al. 2020). Two headwaters on the TP, i.e., HYR and HHR, were selected as the domain of the hydrological modeling (Fig. 1). Six evaluation indices including Nash–Sutcliffe efficiency coefficient (NSE), logarithm Nash–Sutcliffe efficiency coefficient (LogNSE), CC, BIAS, RMSE, and the overall index (OI) (Table 3) were calculated for comprehensively evaluating the simulated discharge (refer to the supplemental material for more information about all parameters used in the CREST-Snow model).

3) EVALUATION USING THE MC METHOD

It is challenging to directly assess the accuracy of the MSP dataset in areas without gauge observations, and therefore, this study used the MC method developed from the triple collocation (TC) method for evaluation (Pan et al. 2015). Two other satellite-based precipitation datasets, CHIRPS and GPCP V3.1 that were not involved in the merging process were used for the MC evaluation based on Eqs. (4)–(6):

$$d_{it}^2 = \frac{1}{2C_{N-1}^2}\sum_{j\neq k}\theta_{jk}d_{jk}^2, \tag{4}$$

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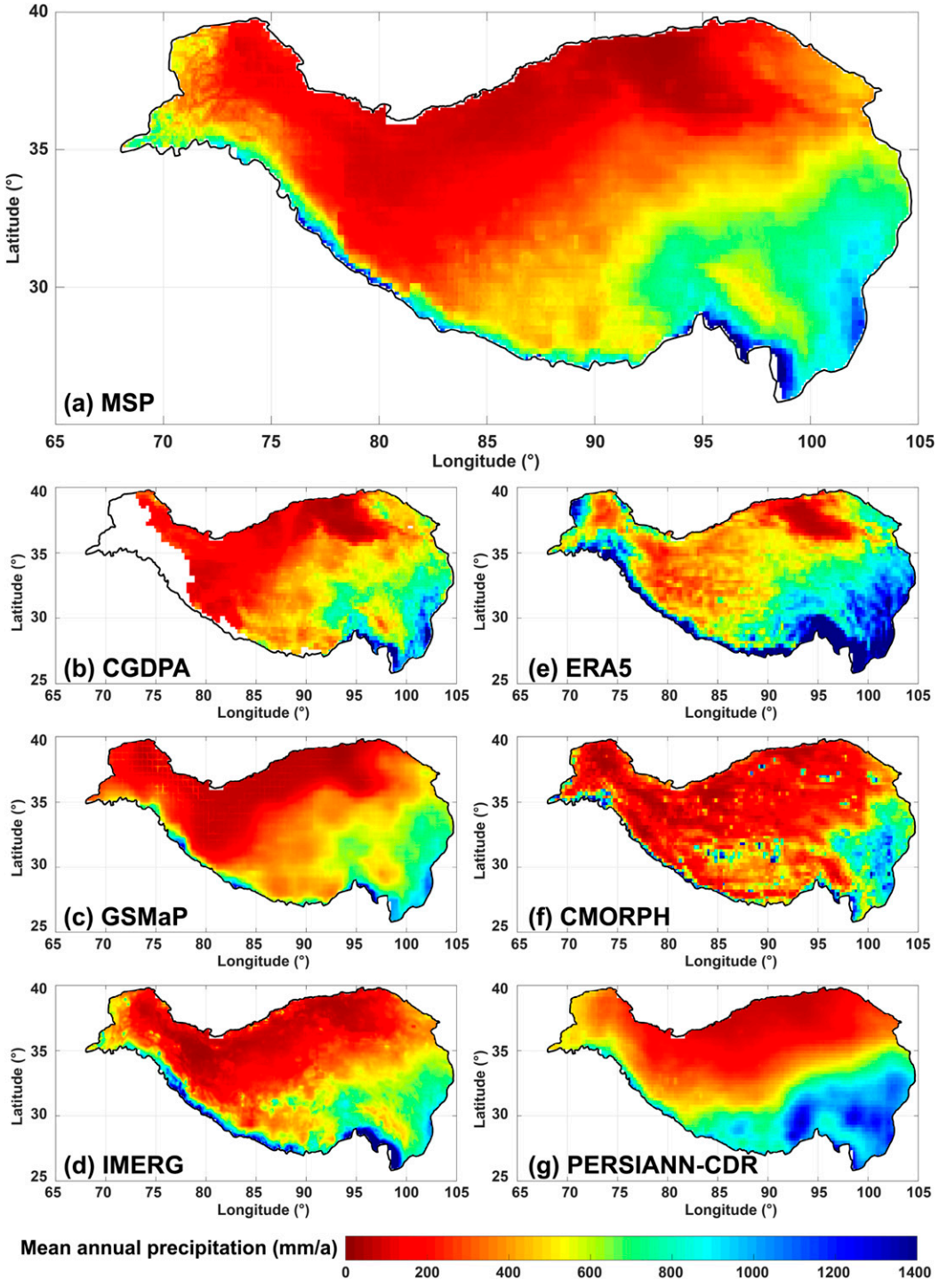


FIG. 6. Spatial distributions of annual precipitation estimated from different products during 2001–14, including (a) MSP, (b) CGDPA, (c) GSMaP, (d) IMERG, (e) ERA5, (f) CMORPH, and (g) PERSIANN-CDR.

$$\theta_{jk} = \begin{cases} N-2 & j=i \quad \text{or} \quad k=i \\ -1 & j\neq i \quad \text{and} \quad k\neq i \end{cases}, \tag{5}$$

$$C_N^2 = N(N-1)/2. \tag{6}$$

The mean of d_{it}, the distance between the true value and estimate i from ith dataset, is the evaluation index in the MC method (the RMSE calculated by the MC method, termed MC RMSE hereafter). The variable N is the number of

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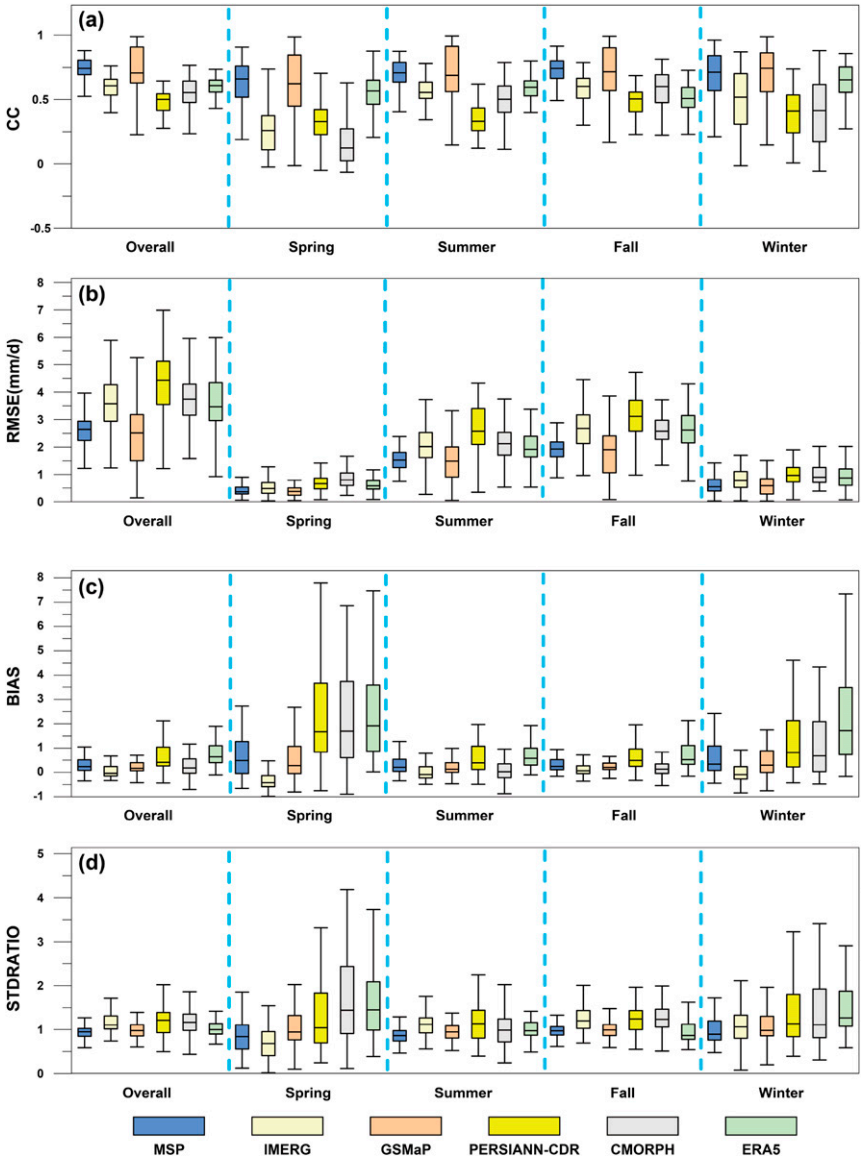


FIG. 7. Boxplots of evaluation metrics of the MSP dataset (blue), IMERG (pale yellow), GSMaP (orange), PERSIANN-CDR (yellow), CMORPH (gray), and ERA5 (green) products based on gauge observations, including: (a) CC, (b) RMSE (mm day^{-1}), (c) BIAS, and (d) STDRATIO. All metrics are shown over the entire study period (Overall) and in spring (March–May), summer (June–August), fall (September–November), and winter (December–February) during 2015–17.

precipitation datasets ($N = 5$ in this study). The distance between estimate j from jth dataset and estimate k from kth dataset is d_{jk}. The two intermediate variables, θ_{jk} and C_N^2, are calculated based on Eqs. (5) and (6).

4. Results

a. Annual precipitation estimation during 1998–2017

Annual precipitation ranges from 350 to 550 mm according to the MSP and five satellite-based products across the TP during 1998–2017 (Fig. 5a). Over the southeast TP (Region I) (see Fig. 1), precipitation estimated from the satellite products shows consistent annual variations and has a similar magnitude except for CMORPH (Fig. 5b), which may be attributed to the accurate estimation of satellite-based products in areas with moderate precipitation (Beck et al. 2017a). CMORPH tends to underestimate precipitation in the central TP and shows much lower values before 2007 compared with other products (Tong et al. 2014a). As for the northeast TP (Region II) (see Fig. 1), annual precipitation ranges from 150 and 350 mm according to the satellite-based estimation; however, interannual variability of different satellite products is not as consistent as that shown

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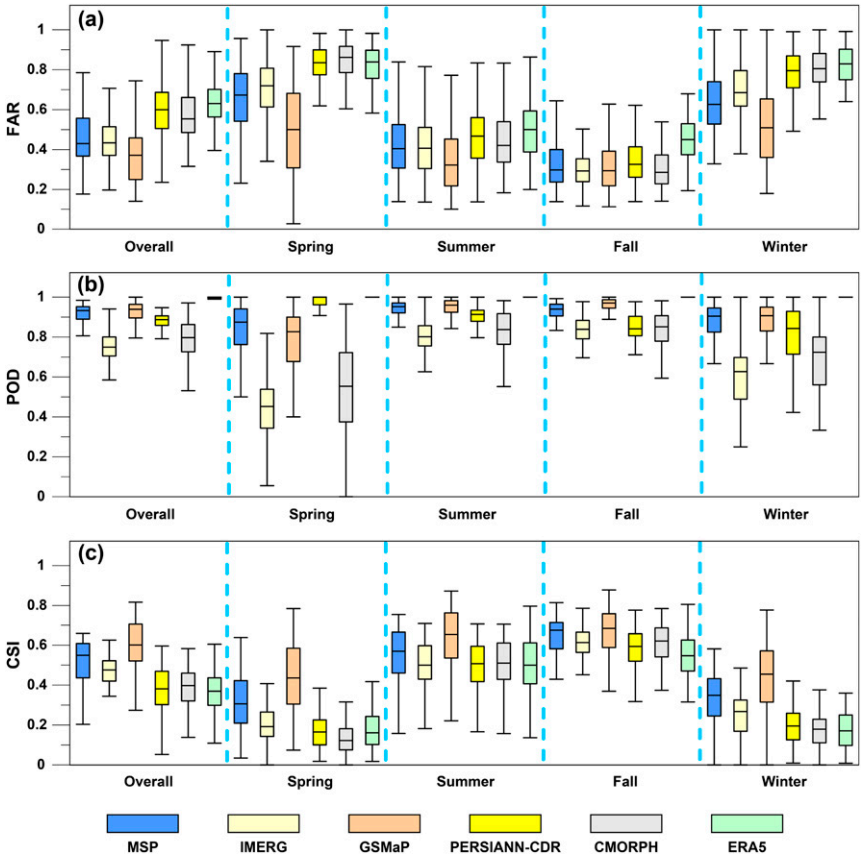


FIG. 8. Detection of rainfall events of the MSP dataset (blue), IMERG (pale yellow), GSMaP (orange), PERSIANN-CDR (yellow), CMORPH (gray), and ERA5 (green) products: (a) false alarm ratio, (b) probability of detection, and (c) critical success index. The evaluation periods and setting of boxplots are the same as Fig. 7.

in Region I (Figs. 5b,c), which may be caused by the poor performance of satellite remote sensing in light precipitation detection (Gao and Liu 2013; Tang et al. 2016a). Over the western TP with sparsely distributed gauges (Region III) (see Fig. 1), annual precipitation ranges from 200 to 600 mm (Fig. 5d). The difference in various precipitation products is larger than that in other two regions, indicating that with the decrease in gauge density, uncertainty in satellite-based precipitation products increases because these products rely on gauge data for correction. In addition, complex topography and low temperatures lead to higher uncertainty in precipitation estimation as well. ERA5 shows more precipitation than other products in three regions, but tends to overestimate precipitation in the TP, with its interannual variability similar to other satellite products. During the 1998–2017 period, mean annual precipitation of the TP was declining at a rate of $0.32\,\text{mm}\,\text{yr}^{-2}$ ($p < 0.05$) based on the MSP dataset (Fig. 5a), whereas mean annual precipitation in Region II (Fig. 5c) shows a strong increase ($2.07\,\text{mm}\,\text{yr}^{-2}$, $p < 0.05$).

As for spatial distribution, annual precipitation shows a decreasing trend from southeast to northwest (Fig. 6). All precipitation products show similar spatial patterns, whereas the MSP dataset has higher estimates than GSMaP, PERSIANN-CDR, and CMORPH on the fringe of the western TP due to the contribution of ERA5 estimation (Figs. 6a,e). Both GSMaP and PERSIANN-CDR products are spatially consistent (Figs. 6c,g), whereas the CMORPH product shows some spatial discontinuity (Fig. 6f). As for IMERG, it shows a similar spatial pattern with MSP, but higher values in the south of the western TP and lower values in the north of the western TP than those of MSP. Compared with other precipitation products, the reanalysis-based ERA5 dataset overestimates precipitation quite a lot, particularly in the southeast region (Fig. 6e).

b. Evaluation based on gauge observations

All precipitation estimates were evaluated based on gauge observations from 2015 to 2017, and six products (i.e., MSP, IMERG, GSMaP, PERSIANN-CDR, CMORPH, and ERA5) were selected for further analysis (Figs. 7 and 8). The CGDPA product is a gauge-based precipitation dataset, so it was excluded from this evaluation. According to the evaluation results (Table 4), the MSP dataset has the highest correlation coefficient with gauge observations (CC = 0.74), the second best RMSE ($2.64\,\text{mm}\,\text{day}^{-1}$) and the third best STDRATIO (0.95). GSMaP has a comparable performance with MSP according to the evaluation of CC (0.71), RMSE ($2.51\,\text{mm}\,\text{day}^{-1}$),

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TABLE 4. First quartile (Q1), median (MED), and third quartile (Q3) of CC, RMSE, BIAS, STDRATIO, FAR, POD, and CSI of six precipitation products (MSP, IMERG, GSMaP, PERSIANN-CDR, CMORPH, and ERA5) during 2015–17. Bold values indicate the best one for each row for comparison among these products.

Statistical metrics		MSP	IMERG	GSMaP	PERSIANN-CDR	CMORPH	ERA5
CC	Q1	**0.69**	0.53	0.63	0.42	0.48	0.56
	MED	**0.74**	0.61	0.71	0.50	0.55	0.61
	Q3	0.81	0.66	**0.91**	0.54	0.64	0.65
RMSE (mm day^{-1})	Q1	2.24	2.94	**1.50**	3.55	3.16	2.96
	MED	2.64	3.57	**2.51**	4.43	3.74	3.47
	Q3	**2.94**	4.27	3.18	5.12	4.29	4.34
BIAS	Q1	0.08	−0.15	0.06	0.26	**−0.04**	0.40
	MED	0.23	**−0.04**	0.16	0.41	0.18	0.64
	Q3	0.51	**0.24**	0.40	1.04	0.55	1.10
STDRATIO	Q1	0.84	**1.01**	0.85	0.93	0.98	0.90
	MED	0.95	1.11	0.98	1.21	1.16	**1.00**
	Q3	**1.03**	1.31	1.11	1.39	1.35	1.13
FAR	Q1	0.37	0.37	**0.25**	0.51	0.49	0.56
	MED	0.43	0.43	**0.37**	0.60	0.55	0.63
	Q3	0.56	0.51	**0.46**	0.69	0.66	0.70
POD	Q1	0.89	0.71	0.90	0.86	0.73	**1.00**
	MED	0.93	0.75	0.94	0.89	0.80	**1.00**
	Q3	0.95	0.80	0.96	0.91	0.86	**1.00**
CSI	Q1	0.44	0.42	**0.52**	0.30	0.32	0.30
	MED	0.55	0.48	**0.60**	0.38	0.40	0.37
	Q3	0.61	0.52	**0.71**	0.47	0.46	0.44

and BIAS (0.16), whereas evaluation of the MSP dataset has the narrowest range of boxplots in most cases, particularly in the evaluation of CC, RMSE, and STDRATIO during the overall period, summer, and fall (Fig. 7). This indicates that the MSP dataset has consistent and stable evaluation results in different geographical areas, which is one of the advantages of MSP. Detection of rainfall events for various precipitation products was compared in terms of the FAR, POD, and CSI (Table 4 and Fig. 8). The MSP dataset has the second-lowest FAR (0.43), the second-highest CSI (0.55), and the third-highest POD (0.93), showing its accuracy in detection of precipitation events. It is worth mentioning that the ERA5 dataset has the highest POD (1.00) and FAR (0.63), which is a possible reason that ERA5 tends to overestimate precipitation in the TP.

Figure 9 shows spatial maps of CC and NRMSE values of the MSP dataset at each gauge in each season. CC values at some gauges are relatively low during spring (e.g., in southwestern regions) and winter (e.g., in south and north regions). In summer and fall, CC values are above 0.65 in most grid cells (>50%) and they are below 0.5 only in several gauges in central and western regions. The overall performance of NRMSE is good, where most grid cells have NRMSE values below 0.1% and ~50% of them are below 0.04. Precipitation is relatively small during spring and winter, and satellite-based precipitation products may overestimate light rain (Gao and Liu 2013; Jiang et al. 2016; Tang et al. 2016a) and have large uncertainty in ice or snow-covered areas (Beck et al. 2017a). This could result in worse evaluation performance in winter and spring than that in summer and fall. Overall, the MSP dataset shows good performance according to the evaluation based on gauge observations, and is reliable particularly in summer and fall. As for the estimation accuracy in winter and spring when precipitation is low, the MSP dataset has similar problems with satellite-based products.

c. *Evaluation using the CREST-Snow model*

In this study, four precipitation datasets having good performance based on gauge data, including MSP, GSMaP, IMERG, and CGDPA, were selected as forcing data of the CREST-Snow model (referring to sections 2b and 4b) to further evaluate their accuracy. Using different precipitation estimates as forcing data, the simulated discharge was compared with discharge observations during 2004–14 in the HYR and HHR basins (Table 5). In the HYR basin, the discharge simulated by the MSP and GSMaP forcing data showed comparable performance with high NSE (NSE = 0.85 for MSP-derived discharge and 0.87 for GSMaP-derived discharge) and LogNSE (LogNSE = 0.90 for MSP-derived discharge and LogNSE = 0.87 for GSMaP derived discharge). The simulated discharge from IMERG precipitation estimates performed slightly worse (NSE = 0.81) than that from the MSP estimates, and the simulated discharge from the CGDPA forcing dataset has the lowest NSE of 0.66. The OI value of the MSP dataset (0.84) in the validation period is lower than that of GSMaP (0.96), and the CGDPA has the lowest OI (0.44). Poor performance of the CGDPA gauge-based product can be attributed to sparse gauge distribution in the HYR basin (fewer than 10 weather stations). In addition, good performance of the simulated discharge from satellite-based products indicates potential applications of remote sensing in hydrological modeling in poorly gauged regions (Sheffield et al. 2018).

In the HHR basin, both MSP and CGDPA forcing datasets performed much better than other precipitation products with high NSE (NSE = 0.81 for MSP-derived discharge and 0.83 for

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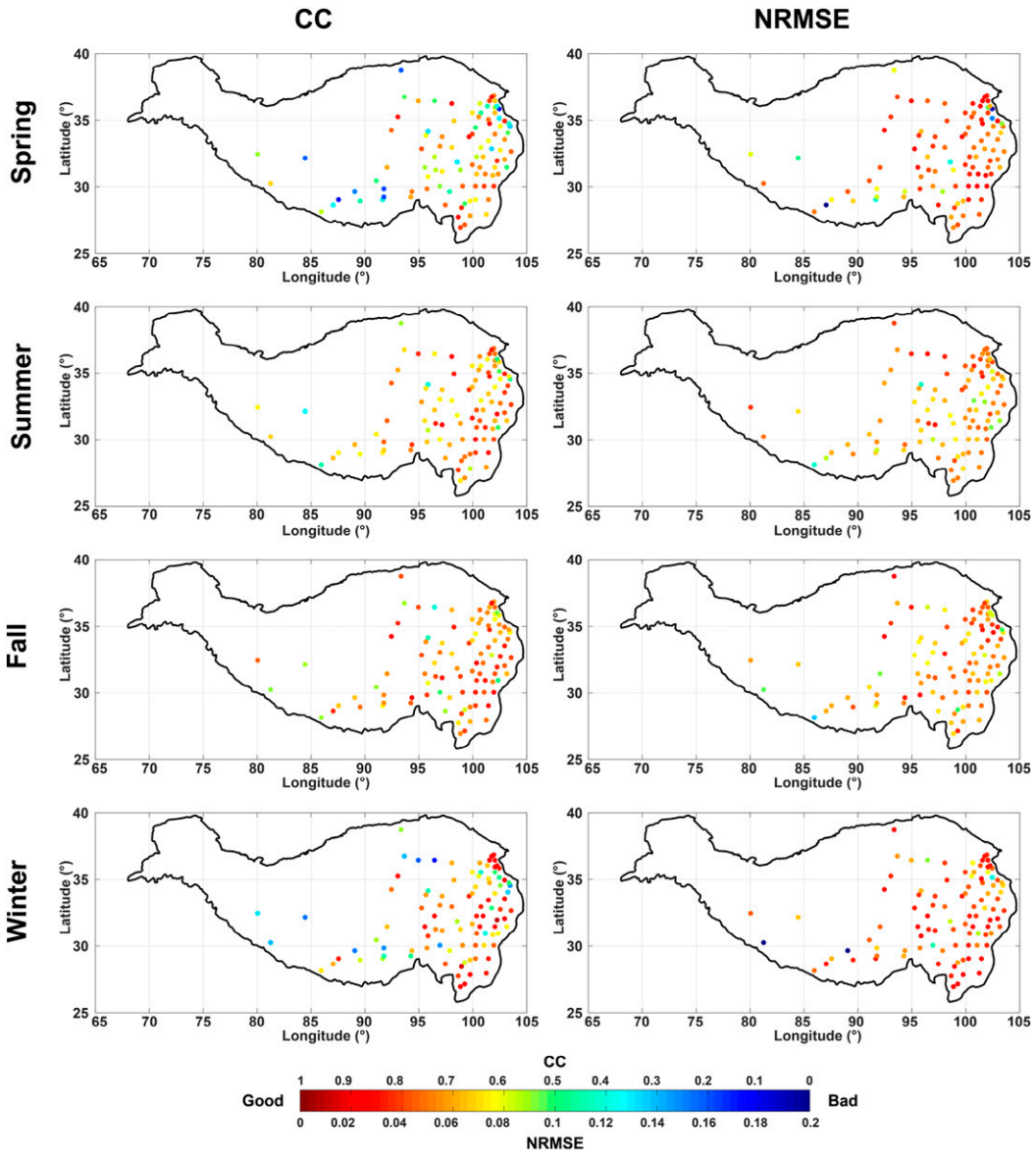


FIG. 9. (left) CC and (right) NRMSE values of the MSP dataset at each rain gauge in spring, summer, fall, and winter during 2015–17.

CGDPA-derived discharge) and LogNSE (LogNSE = 0.84 for MSP-derived discharge and 0.83 for CGDPA-derived discharge). GSMaP-derived discharge had good performance in the HYR basin but poor performance in the HHR basin with an NSE of 0.69. IMERG-derived discharge had slightly worse performance in the HHR basin with an NSE of 0.69 than that derived from GSMaP. As for the OI values of the validation period, the MSP dataset is the best with an OI of 0.88, which outperforms other products.

Here we particularly show discharge simulations driven by the MSP dataset for the 2004–14 period (Fig. 10). Peak values of the discharge simulations are consistent with the observed values in both HYR and HHR basins. The discharge simulation during the low-flow period in the HYR basin is better than that in the HHR basin that shows overestimation. Overall, the CREST-Snow model using MSP as the forcing dataset can well simulate discharge in both the HYR and HHR basins, indicating high quality of the MSP precipitation estimates.

d. Evaluation based on the multiple collocation (MC) method

Based on the evaluation results in sections 4b and 4c, here we compare two high-quality precipitation products, i.e., GSMaP and MSP, at the monthly time scale during the 2003–12 period using the MC method (Fig. 11). In addition, the ERA5, GPCP V3.1, and CHIRPS products were involved in the MC method, and the GPCP V3.1 and CHIRPS estimates were independent of the merged MSP dataset. The reanalysis-based

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TABLE 5. Indirect evaluation of precipitation estimates for the HYR and HHR basins using the CREST-Snow model. Information and formulas of calculating these indices can refer to Table 3. The best evaluation results for each metric are bolded.

Basin		HYR				HHR			
Precipitation		CGDPA	IMERG	GSMaP	MSP	CGDPA	IMERG	GSMaP	MSP
Calibration (2009–14)	NSE	0.80	0.79	**0.87**	0.84	**0.86**	0.66	0.78	0.81
	LogNSE	0.81	0.75	0.88	**0.90**	0.77	0.73	0.80	**0.84**
	CC	0.91	0.90	**0.94**	0.92	**0.92**	0.84	0.91	0.91
	BIAS	**1.6**	−46.5	−9.0	−1.9	17.7	−80.3	**−2.3**	56.3
	RMSE	278	279	220	**206**	216	335	266	**202**
	OI	0.68	0.40	0.84	**0.88**	**0.84**	0.40	0.76	**0.84**
Validation (2004–08)	NSE	0.41	0.82	**0.87**	0.86	0.77	0.68	0.50	**0.81**
	LogNSE	0.75	0.76	0.87	**0.89**	0.82	0.79	0.80	**0.84**
	CC	0.86	0.90	**0.93**	**0.93**	**0.91**	0.86	0.88	0.90
	BIAS	87.3	−42.4	**21.7**	−50.9	95.3	**22.0**	111.1	−24.6
	RMSE	395	217	**197**	234	**225**	266	330	231
	OI	0.40	0.68	**0.96**	0.80	0.84	0.60	0.48	**0.88**
Overall (2004–14)	NSE	0.66	0.81	**0.87**	0.85	**0.83**	0.67	0.69	0.81
	LogNSE	0.78	0.79	0.87	**0.90**	0.83	0.79	0.80	**0.84**
	CC	0.88	0.91	**0.93**	0.92	**0.91**	0.84	0.89	0.90
	BIAS	40.6	−44.7	**4.9**	−28.6	53.0	−33.8	48.2	**12.2**
	RMSE	336	253	**210**	222	**220**	305	297	253
	OI	0.44	0.56	**0.96**	0.84	0.84	0.48	0.60	**0.88**

ERA5 was taken into consideration because it is more accurate in high-altitude and snow-covered areas (Beck et al. 2017a; Tang et al. 2018a). The results of MC RMSE maps of MSP and GSMaP are shown in Fig. 11, indicating that the mean MC RMSE values over the TP for the MSP and GSMaP estimates are 0.28 and 0.35 mm month^{-1}, respectively. MC RMSE values of both MSP and GSMaP datasets are low in the eastern TP (Region I) that has dense gauge observations. In the poorly gauged western TP (Region III), MC RMSE values of these two products show large differences (the maximum $<$ 1.4 mm month^{-1} for the MSP dataset but near 2 mm month^{-1} for the GSMaP dataset). In addition, MC RMSE of the MSP dataset in the western TP (0.31 mm month^{-1}) is close to that over the TP (0.28 mm month^{-1}), whereas MC RMSE of the GSMaP dataset in the western TP is 0.44 mm month^{-1}, 26% larger than that over the TP (0.35 mm month^{-1}).

These results indicate that the MSP dataset improves the accuracy of precipitation estimation over the TP, particularly in poorly gauged regions, a significant advantage of multisource data merging over other single datasets (refer to the supplemental material for more comparison results by the MC method).

5. Discussion

a. *Advantages of the merging methodology*

Environmental variables including elevation, temperature, air pressure, and wind speed were involved in deriving weights of various precipitation products. The ANN method determines weights from the relationship between the quality of products and environmental variables and therefore, the advantages of different datasets are maximized. The weight calculation method does not depend on gauge density and then the determination of weights for each grid cell is fair and independent, which is completely different from other interpolation methods. In fact, it is hard to directly quantify the relationship between environmental conditions and the quality of precipitation products and consequently, the ANN method is a promising approach of capturing the complex relationship between inputs and outputs. Beck et al. (2017a) used an interpolation approach to generate continuous weights of the MSWEP product, and reanalysis-based datasets are the dominating component in the TP according to their setting of temperature thresholds. However, the reanalysis product, ERA5, overestimates precipitation compared with other products in the TP (referring to section 4a), which may result in uncertainty in the merged MSWEP product.

In addition, weights from widely used spatial interpolation methods are not reliable in such a poorly gauged region (Xie et al. 2007). Ma et al. (2018) used a dynamic Bayesian model averaging (BMA) scheme to merge multisatellite precipitation over the TP. The BMA method directly considers the statisticalrelationship between satellite data and precipitation observations, and the estimation accuracy is influenced mostly by environmental conditions (e.g., elevation) according to their evaluation results. The RMSE and relative bias of BMA-derived precipitation estimates are higher than those of the CMORPH product when the elevation is over 3000 m (Ma et al. 2018). In our study, BIAS of the MSP (0.23) and CMORPH (0.18) datasets are similar and the MSP estimates have much smaller RMSE (2.64 mm day^{-1}) than the CMORPH product (3.74 mm day^{-1}) (see section 4b). In addition, the MSP dataset has high quality over the TP because the distribution of RMSE does not change sharply (see section 4d), but the BMA-derived estimates have large uncertainty at high elevations (Ma et al. 2018). Selection of environmental variables can be changed in other study regions to achieve better results, and other

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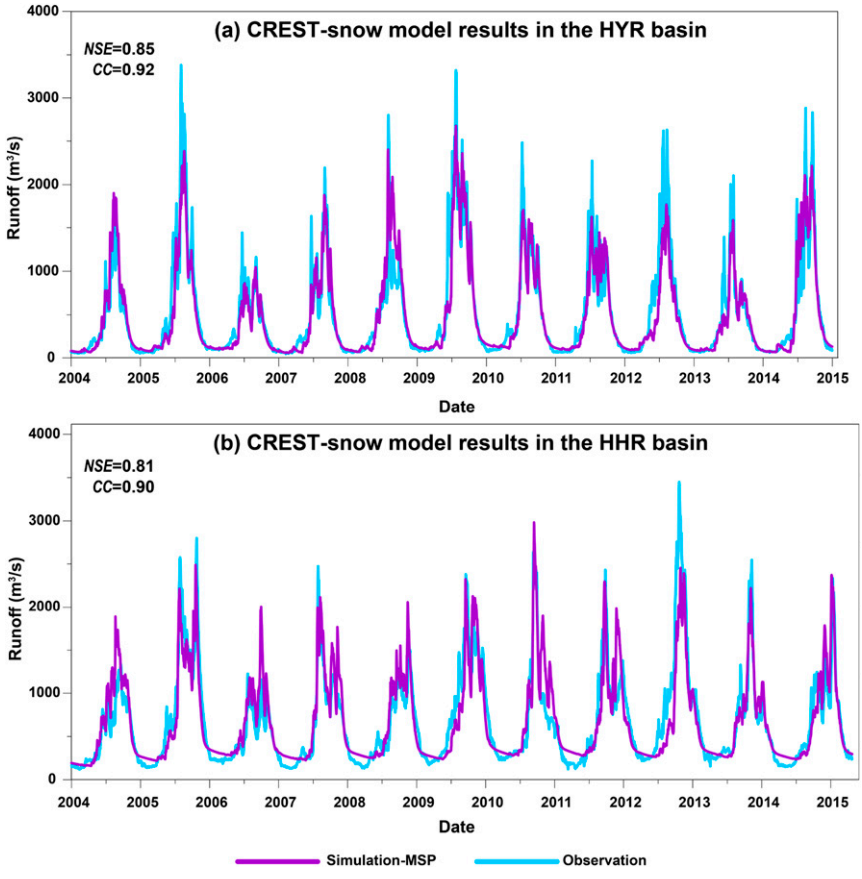


FIG. 10. Discharge observations (blue line) and simulations at the daily time scale driven by the MSP dataset generated in this study (purple line) in the (a) HYR and (b) HHR basins.

machine leaning algorithms (e.g., random forest and deep neural networks) could be used to determine the weights. Overall, results of our study provide reference for merging multisource precipitation data.

b. *Evaluation approaches*

This study used three different approaches to fully evaluate precipitation products, including the evaluation based on gauge observations, hydrological modeling, and the MC method. However, some issues need to be well considered. First, as for the evaluation method based on gauge observations, all gridded precipitation products show the mean estimate in each grid cell, which is different from the gauge-based precipitation measurements in nature at the point scale, particularly in areas with complex terrain and sparse stations such as the TP. The difference in spatial scale between satellite or reanalysis-based precipitation estimation and gauge-based precipitation measurements is one source of uncertainty in evaluation. Second, as for the indirect assessment using hydrological modeling, performance of discharge simulation may be affected by not only the quality of the primary forcing datasets, but also other factors (e.g., model structure and parameters). The CREST-Snow model has been successfully applied to discharge simulations in the TP (Chen et al. 2017; P. Han et al. 2019; Z. Han et al. 2019; Huang et al. 2020), whereas in other areas and studies, the applicability of the model needs to be fully examined. Third, datasets evaluated by the MC method should theoretically be independent to ensure the calculation results. The five datasets (i.e., MSP, GSMaP, ERA5, CHIRPS, and GPCP) were used for the MC assessment, but the three satellite-based precipitation datasets (i.e., GSMaP, CHIRPS, and GPCP) are not completely independent, because they are corrected using gauge observations. The correction by gauge observations has almost no effect in Region III (see Fig. 1) with a sparse gauge distribution, and therefore, the three satellite-based products are thought to be independent in this area. The results of the MC evaluation is much more reliable in Region III than other regions due to the independence of the precipitation products. According to the evaluation results of the MC method, the MSP dataset performs best over the entire TP, particularly in the western TP (Region III).

c. *Future development and application*

This study developed a promising approach based on the ANN method to merge precipitation datasets, aiming to provide reference for hydrometeorological studies. In addition, with the development of satellite retrieval algorithms and

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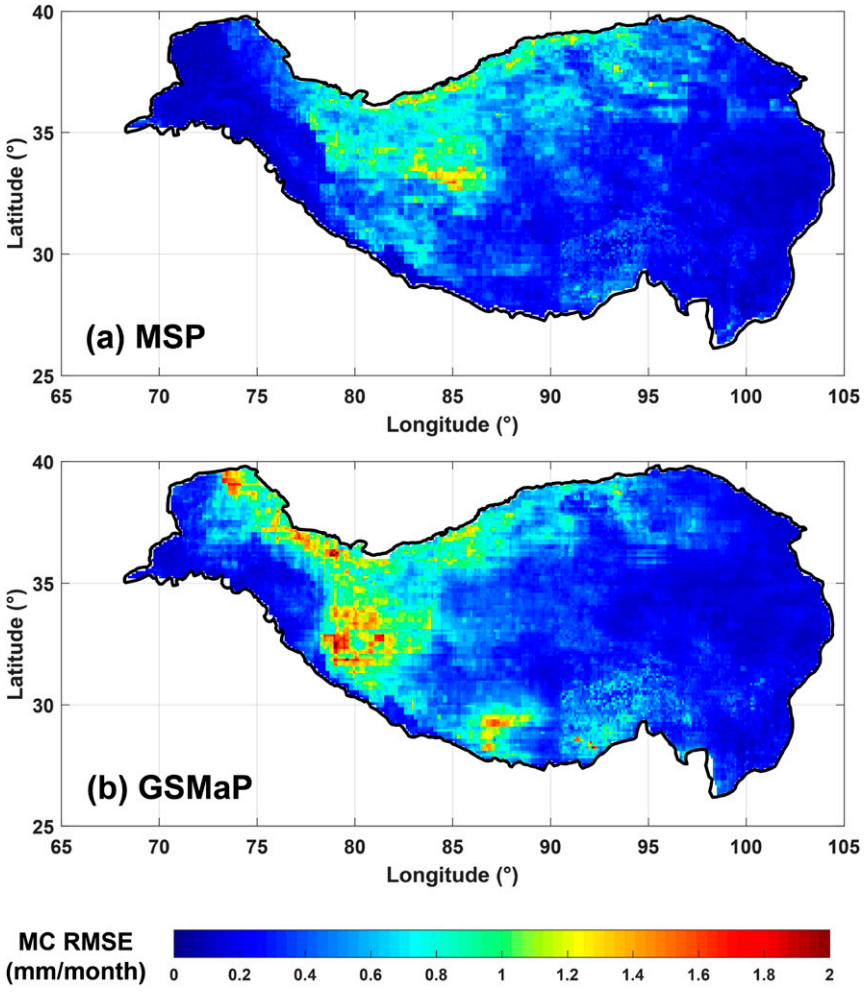


FIG. 11. Maps of MC RMSE values of (a) MSP and (b) GSMaP based on the MC evaluation method over the TP at the monthly scale during 2003–12.

reanalysis products, precipitation products keep evolving based on the merging method, e.g., IMERGV06 and ERA5 are the new versions of TRMM and ERA-Interim, respectively (Hoffmann et al. 2019; Hou et al. 2014; Tang et al. 2016b; C. Wang et al. 2019). Machine learning algorithms developed in recent years are widely used in precipitation research, such as using convolutional neural networks to correct for error and improve precipitation estimation (Pan et al. 2019), using random forests for spatial downscaling (He et al. 2016), and using deep neural networks to identify rain and snow phases (Tang et al. 2018a). These studies showed great potential of machine learning in hydrological applications. Through the learning process, the complex relationship between input and output data can be simulated, and it is continuously improved to obtain the logical or mathematical relationship implicit in each variable. It has made a significant contribution to capturing the internal relationship of various hydrological elements.

There are currently still some issues associated with the application of machine learning to be resolved, including 1) a large number of accurate training data are required to ensure excellent learning results, 2) some simple and direct methods may be better than machine learning if the relationships are obvious among variables, and 3) the learning process may not conform to the physical or hydrological rules in some cases. In general, extending the length of data time series may be a practical way to explore sufficient gauge data as the training set in data-scarce areas, and abnormal values should be eliminated to avoid the adverse effect on the learning results. Furthermore, for areas having reliable and abundant data for training, some straightforward methods (e.g., linear regression and spatial interpolation) may perform better than machine learning. Overall, the progress in different fields should complement each other to achieve better results, and machine learning methods would show further development in hydrological studies.

Methods developed in this study have potential to be applied to other poorly gauged regions. Precipitation datasets and environmental variables should be carefully selected for different regions with different geographical and environmental settings, given that the impact of environment variables on the

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quality of precipitation estimates could be region specific. The generated MSP dataset is valuable in hydrometeorological research and applications including spring or summer flood mitigation for the TP and downstream areas.

6. Conclusions

We generate the MSP precipitation dataset at a daily time scale and a spatial resolution of 0.1° across the TP over the 1998–2017 period. Multisource precipitation datasets, including five satellite remote sensing products, two gauge-based products, and one reanalysis dataset are optimally merged using the ANN method with the consideration of the impact of environmental variables. The MSP dataset provides reliable precipitation estimates over the TP with high quality and reasonable spatiotemporal variability. Primary advantages of the MSP dataset include 1) high-quality and high spatiotemporal resolution in poorly gauged regions, 2) generally consistent performance in space and time, and 3) the potential of incorporating more high-quality precipitation products in the merging process.

Some limitations of the MSP dataset will be addressed, including 1) false detection for nonprecipitation events in some cases, 2) relatively low accuracy of estimation for light precipitation, particularly during spring and winter seasons over snow-cover regions, and 3) uncertainty in deriving weights using the ANNs. Overall, the method shows the potential of machine learning in optimizing various precipitation sources over poorly gauged regions. The generated MSP dataset could be considered a substitute for ground observations across the TP and provide valuable information on precipitation input for hydrometeorological research and applications.

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